Samuel A. Ballas

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Employment

Florida State University — Tallahassee, Florida

- Director of Pure Mathematics — 2023-Present
- Associate Professor — 2022-Present
- Assistant Professor — 2016-2022

University of California at Santa Barbara — Santa Barbara, California

- RTG Visiting Assistant Professor — July 2013-July 2016

Education

University of Texas at Austin — Austin, Texas

- Ph.D., Mathematics — May 2013
  Thesis: *Flexibility and Rigidity of Three-Dimensional Convex Projective Structures.*
  Advisor: Alan Reid
- M.A., Mathematics — May 2011

Emory University — Atlanta, Georgia

- B.S. Mathematics., *Summa cum Laude* — May 2007
  Thesis: *Geometrically Finite Fuchsian Groups and the Patterson-Sullivan Measure.*
  Advisor: David Borthwick

Research Interests

Low-dimensional topology, hyperbolic geometry, real projective geometry, discrete groups, deformation theory, higher Teichmüller theory.

Publications and Preprints

- *Parabolic-preserving deformations of cusped hyperbolic lattices* (with J. Paupert and P. Will), arXiv:2605.03161
- *Geometric Perspective on Concentration Phenomena in Frame Theory* (joint with F. Karabatman and T. Needham), arXiv:2605.03867
- *On the existence of Parseval frames for vector bundles*
  (with T. Needham and C. Shonkwiler), Trans. Amer. Math. Soc. Ser. B 12 (2025), 395-416
- *Tame and relatively elliptic $\mathbb{CP}^1$-structures on the thrice punctured sphere*
  (with P. Bowers, A. Casella, and L. Ruffoni), Alg. Geom. Top. 24-8 (2024), 4589-4650
- *The moduli space of marked generalized cusps in real projective manifolds*
  (with D. Cooper and A. Leitner), Conf. Geom. & Dyn. vol 26 (2022), 111-164
- *Gluing equations for real projective structures on 3-manifolds*
  (with A. Casella), Geom. Dedicata 215 (2021) 69-131.

- *Thin subgroups isomorphic to Gromov–Piatetski-Shapiro lattices*
  Pacific J. Math. 309 (2020), no 2, 257–266.
- *Constructing thin subgroups of* $\mathrm{SL}(n+1,\mathbb{R})$ *via bending*
  (with D.D. Long), Algebr. Geom. Topol. 20-4 (2020), 2071–2093
- *Constructing convex projective 3-manifolds with generalized cusps*
  J. London. Math. Soc. 103-4 (2021) 1276–1313.
- *Generalized cusps in real projective manifolds: classification*
  (with D. Cooper and A. Leitner), J. Topol. 13-4 (2020) 1455–1496
- *Properly convex bending of hyperbolic manifolds*
  (with L. Marquis), Groups, Geom, & Dynam 14 (2020), no. 2, 653-688.
- *Convex projective structures on non-hyperbolic 3-manifolds*
  (with J. Danciger and G.S. Lee), Geom. & Topol 22 (2018) 1593–1646.
- *Constructing thin subgroups commensurable with the figure-eight knot group*
  (with D.D. Long), Algebr. Geom. Topol. 15 (2015) no. 5, 3011-3024.
- *Finite volume properly convex deformations of the figure-eight knot*,
  Geom. Dedicata 178 (2015), 49-73.
- *Deformations of non-compact, projective manifolds*,
  Algebr. Geom. Topol. 14 (2014), no. 5, 2595-2625.

In Preparation

- *Convex projective Dehn filling*
  (with J. Danciger, G.S. Lee, and L. Marquis).

Selected Talks

- *Parseval Frames on Vector Bundles*
  AMS Southeast Sectional Meeting, Mar 2026
- *Parseval frames on vector bundles*
  Codes and Expansions (CodEx) Seminar, Online seminar Feb 2024
- *A convex projective Dehn filling theorem*
  AMS Southwest Sectional Meeting, GA Tech Mar 2023
- *Complex Projective Structures on Surfaces*
  University of Florida, Colloquium Feb 2022
- *Properly Convex Dehn Fillings*
  Virginia Geometry Seminar Oct 2021
- *Triangulations in Low Dimensional Geometry and Topology*
  Cal Poly San Luis Obispo, Colloquium Feb 2021
- *Gluing Equations for Projective Structures on 3-manifolds*
  Oklahoma State University, Topology Seminar Nov 2020
- *Gluing Equations for Projective Structures on 3-manifolds*
  Trends in Low Dimensional Topology, Online Seminar Jun 2020

- *Gluing Equations for Projective Structures on 3-manifolds*[*]
  Oklahoma State University, Topology Seminar — Apr 2020
- *Gluing Equations for Projective Structures on 3-manifolds*[*]
  AMS Sectional Meeting, Special Session on Trends in Teichmüller Theory — Mar 2020
- *Gluing Equations for Projective Structures on 3-manifolds*
  FRG Workshop, University of Michigan — Dec 2019
- *Gluing Equations for Projective Structures on 3-manifolds*
  Geometry Seminar
  Stanford University — Mar 2019
- *Thin Groups via Bending*
  Topology Seminar
  Rice University — Feb 2019
- *Thin Groups from Projective Geometry*
  Thin Groups in Number Theory, Geometry and Topology
  Rice University — May 2018
- *Convex Projective Structures in Low Dimensions*
  University of Georgia, Topology Seminar — April 2018
- *Generalized Cusps in Convex Projective Manifolds*
  GEAR Retreat, Stanford University — August 2017
- *Generalized Cusps in Convex Projective Manifolds*
  Dynamics on Character Varieties Workshop, University of Rennes I — June 2017
- *Rank 1 Deformations of Non-cocompact Hyperbolic Lattices*
  University of Texas Topology Seminar — March 2017
- *Geometric Structures on Manifolds*
  UF/FSU Topology and Dynamics Meeting
  University of Florida — February 2017
- *Classification of Generalized Cusps*
  Joint Mathematics Meetings, Special Session on Group Actions and Geometric Structures, Atlanta, GA — January 2017
- *Classification of Generalized Cusps*
  California Institute of Technology, Geometry and Topology Seminar — April 2016
- *Geometric Structures on Manifolds*
  Mathematics Colloquium
  Florida State University — January 2016
- *Gluing Convex Projective Manifolds*
  Higher Teichmuler theory and Higgs bundles: interactions and new trends
  Heidelberg — November 2015

[*]Cancelled due to COVID-19 pandemic

- *Convex Projective Structures on Manifolds*
  A Mini Course at Workshop on Geometric Structures
  KIAS, Seoul, South Korea November 2014
- *Geometry of Cusps in Convex Projective Manifolds*
  Workshop on Geometry and Physics, Univeristy of Pittsburgh May 2014
- *Geometry of Cusps in Convex Projective Manifolds*
  RTG Workshop on Geometric Structures and Discrete Groups, UT Austin May 2014
- *A Geometric Description of (P)*$SL_4(\mathbb{R})$ *Hitchin Representations*
  Workshop on Higher Teichmüller-Thurston Theory, Northport, Maine June 2013
- *Convex Projective Structures on Non-compact 3-manifolds*
  California Institute of Technology, Geometry and Topology Seminar May 2013
- *Convex Projective Deformations of the Figure-8 Knot Complement*
  AMS Special Session on Algebraic and Geometric Structures of 3-manifolds April 2013
- *Convex Projective Deformations of Non-Compact Manifolds*
  Arizona State University Differential Geometry Seminar January 2013
- *Convex Projective Deformations of Non-Compact Manifolds*
  Purdue University Geometric Analysis Seminar January 2013
- *Convex Projective Structures on 3-manifolds*
  University of California at Santa Barbara Topology Seminar October 2012
- *Twisted Cohomology and Deformation Theory*
  GEAR Junior Retreat, Urbana, Illinois July 2012

Teaching Experience

Florida State University

- MAA 4402 *Complex Analysis* Summer 2026
- MAS 3105 *Linear Algebra with Applications* Spring 2026
- MAC 2313 *Calculus III* Fall 2025
- *MTG 5932, Differential Geometry of Principal Bundles* Summer 2025
- MAC 2313 *Calculus III* Spring 2025
- MGF 3301 *Intro to Advanced Math* Fall 2024
- MTG 5932 *Lie groups* Summer 2024
- MAS 5932 *Arithmetic groups* Spring 2024
- MAA 4402 *Complex Variables* Summer 2023
- MAC 2313 *Calculus III* Spring 2023
- MTG 5327 *Topology II* Spring 2023
- MTG 4212, *College Geometry* Fall 2022
- MTG 5326 Graduate Topology I Fall 2022
- *MTG 5932, Differential Geometry of Principal Bundles* Spring 2022
- *MAD 3105, Discrete Math II* Spring 2022
- *MAC 2313, Calculus III* Fall 2021
- *MTG 5932, Geometric Structures on Manifolds* Spring 2021
- *MAD 3105, Discrete Math II* Fall 2020
- *MAC 2312, Calculus II* Spring 2020

- *MAC 2313, Calculus III* Fall 2019
- *MAC 2312, Calculus II* Spring 2019
- *MTG 5327, Graduate Topology II* Spring 2019
- *MAC 2313, Calculus III* Fall 2018
- *MTG 5326, Graduate Topology I* Fall 2018
- *MAP 2303, ODEs* Spring 2018
- *MAC 2312, Calculus II* Spring 2017
- *MAC 2313, Calculus III* Fall 2016

University of California, Santa Barbara

- *Math 260P, Projective Structures on Manifolds* Spring 2016
- *Math 108A, Introduction to Linear Algebra* Fall 2015
- *Math 8, Transition to Higher Mathematics* Fall 2015
- *Math 260P, The Casson Invariant* Spring 2015
- *Math 6A, Vector Calculus* Winter 2015
- *Math 108A, Introduction to Linear Algebra* Fall 2014
- *Math 260P, Representations of Surface Groups* Spring 2014
- *Math 8, Transition to Higher Mathematics* Winter 2014
- *Math 103, Introduction to Group Theory* Fall 2013

Mentorship

Ph.D. Students

- Jared Miller 2024
- Ferhat Karabatman (joint with T. Needham) Current

Faculty

- Oishee Banerjee (teaching mentor) 2023-Present
- Ali Kara (teaching mentor) 2024-Present

Postdocs

- Wenwen Li (teaching mentor) 2025-Present
- Brandon Doherty (teaching mentor) 2023-Present
- Florian Stecker (supervisor) 2022-2024
- Eunjung Noh (teaching mentor) 2021-2023
- Ignat Soroko (teaching mentor) 2021-2022
- Alex Casella (supervisor) 2018-2021

Ph.D. Committee Member

- Pan Fang (advisor Tom Needham & Washington Mio)
- Mujtaba Ali (advisor Tom Needham)
- Theodore Weisman (advisor Jeffrey Danciger, University of Texas at Austin)
- Paul Eugenio (advisor Oskar Vafek, Physics)
- Anindya Chanda (advisor Sergio Fenley)
- Aamir Rasheed (advisor Wolfgang Heil)
- Jay Leach (advisor Kathleen Petersen)
- Opal Graham (advisor Phil Bowers)
- John Bergschneider (advisor Wolfgang Heil)

- Leona Sparaco (advisor Kathleen Petersen)

Honors in the Major

- Zachary Levy (Committee member, Physics)
- Jessica Bellaire (supervisor) 2024

Independent Study

- Joseph Gerbasio (2026, undergraduate, differential geometry)
- Jessica Bellaire (2022, undergraduate, mathematics of Euler disks)
- Daniel Hartman (2018, graduate student, geometric structures and characteristic classes)
- Jose Bousa (2017, undergraduate, projective geometry)

Funding and Awards

Grants Awarded

- Committee on Faculty Research Support award, Council for Research and Creativity (COFRS), Florida State University, $20,000, Summer 2022.

- Supplement to DMS 1709097, $27,483, 2021-2023.

- NSF research grant DMS 1709097- *Geometry and topology of convex projective manifolds*, $139,278, 2017-2021

- First Year Assistant Professor Award, Council for Research and Creativity, Florida State University, $20,000, Summer 2017

Conferences and Workshops

- AMS Special Session
  AMS Southeastern sectional meeting
  Florida State University Mar 2024
- UF/FSU Topology and Geometry meeting
  Florida State University Nov 2023
- AMS Special Session
  AMS Southeastern sectional meeting
  University of Florida Nov 2019
- UF/FSU Topology and Geometry meeting
  Florida State University Feb 2019
- GEAR network junior retreat
  Stanford University July 2017

Departmental Service

- Executive Committee 2025-Present
- Director of Pure Mathematics 2023-Present
- Faculty Evaluation Committee 2023-Present
- Algebra Tenure Track Hiring Committee 2022-23
- Geometry/Topology Tenure Track Hiring Committee 2021-22
- Organized Job Application Workshop 2019
- Library Committee 2018-22
- Organizer (Topology seminar) 2018-2024

University Service

- Road Scholar Committee 2025-Present
- Faculty Senator 2021-2026

Professional Service

- Journal Referee
  - *Journal of Experimental Mathematics*
  - *Journal of Differential Geometry*
  - *Transactions of the AMS*
  - *Publications Mathématiques de l'IHES*
  - *International Mathematics Research Notices*
  - *Journal of the London Mathematical Society*
  - *Algebraic and Geometric Topology*
  - *Geometriae Dedicata*
  - *Pacific Journal of Mathematics*
  - *Proceedings of the AMS*
- Reviewer for Mathematical Reviews 2014-Present
- Judge for Capital Regional Science and Engineering Fair 2017, 2019, 2024

Honors, Awards, and Fellowships

- Research Training Grant Postdoc, UC Santa Barbara 2013-2016
- Research Training Grant Fellowship, UT Austin 2011-2012
- Deborah Jackson Award recipient, Emory University 2007

Professional Memberships and Associations

- Member of the American Mathematical Society (AMS) 2007-present.

Last updated May 7, 2026